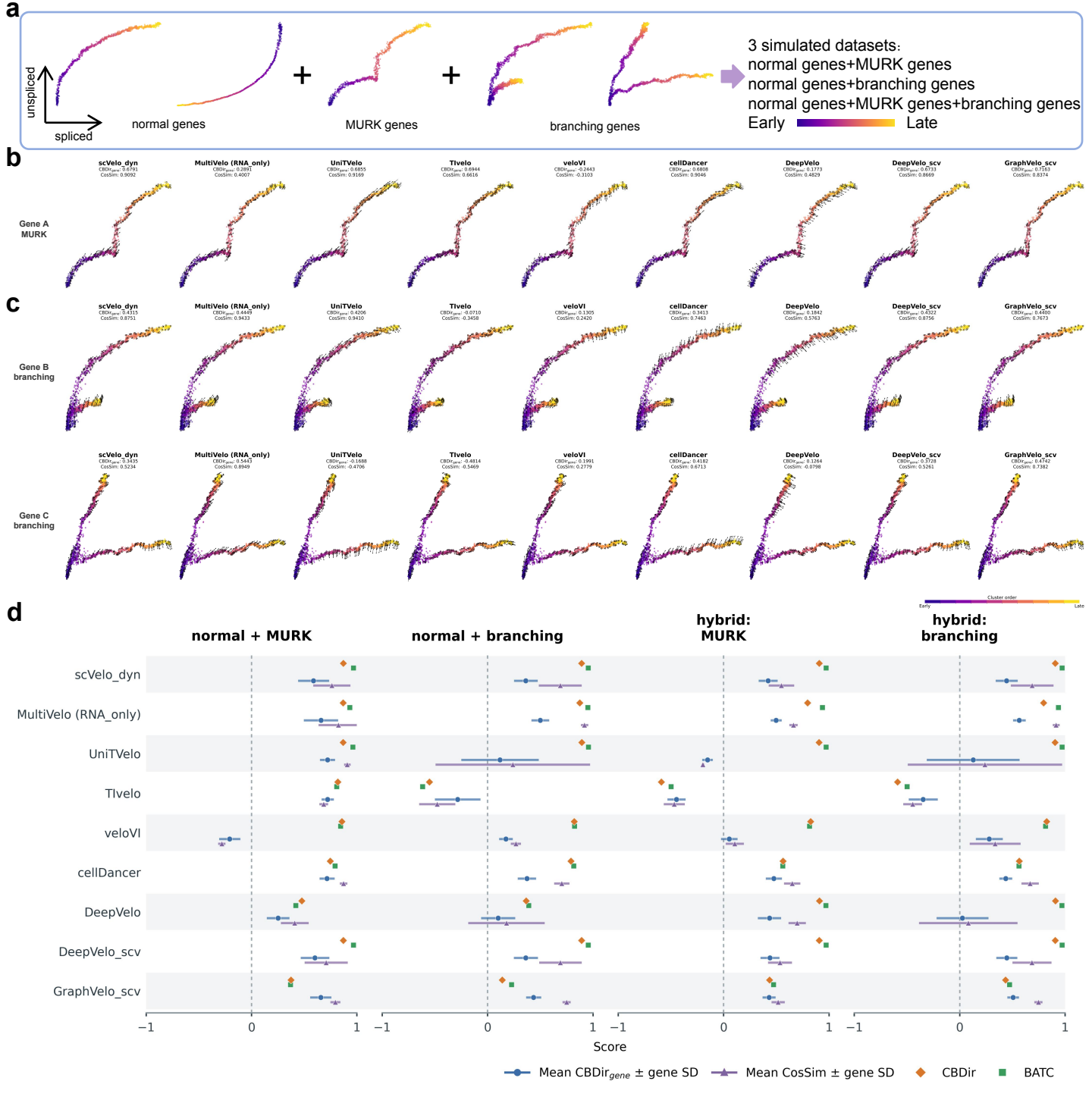

a
unspliced
spliced
normal genes
MURK genes
branching genes
3 simulated datasets:
normal genes+MURK genes
normal genes+branching genes
normal genes+MURK genes+branching genes
Early
Late
b
Gene A
MURK
c
Gene B
branching
Gene C
branching
scVelo_dyn
MultiVelo (RNA_only)
UniTVelo
TIvelo
veloVI
cellDancer
DeepVelo
DeepVelo_scv
GraphVelo_scv
d
normal + MURK
normal + branching
hybrid:
MURK
hybrid:
branching
Cluster order
Early
Late
scVelo_dyn
MultiVelo (RNA_only)
UniTVelo
TIvelo
veloVI
cellDancer
DeepVelo
DeepVelo_scv
GraphVelo_scv
−1
0
1
Score
Mean CBDir$_{gene}$ ± gene SD
Mean CosSim ± gene SD
CBDir
BATC